

Agentic AI Systems

Hands-On Coding Problems

Course: Agentic AI Systems

Module 2 — LLMs as Reasoning Engines:

Project 2 Problem: Prompt Engineering Playground — News Classifier

Topic: LLM Foundations, Tokenisation, Prompt Engineering

Problem Statement

You have a dataset of 50 news headlines (provided as a plain text file, one headline per line). Build a CLI tool that classifies each headline into one of five categories: Politics, Technology, Sports, Finance, Health.

1. Implement zero-shot classification using a single system prompt.
2. Implement few-shot classification by prepending 3 labelled examples per category.
3. Implement Chain-of-Thought classification where the model explains its reasoning before giving the final label.
4. Compare accuracy of all three approaches against a hand-labelled ground truth file. Print a comparison table.
5. Measure and compare average token usage per approach.

Learning Objectives

- Understand the practical difference between zero-shot, few-shot, and CoT prompting
- See how prompt design directly affects output quality and cost
- Parse structured LLM output (extract the label from a reasoning trace)

Starter Hints

```
def classify_zero_shot(headline: str) -> str:
    prompt = f"Classify this headline into one of [Politics, Technology, Sports, Finance, Health]: {headline}"
    ...

def classify_few_shot(headline: str, examples: list[dict]) -> str:
    few_shot_block = "\n".join([f"Headline: {e['text']}\nCategory: {e['label']}" for e in examples])
    ...

def classify_cot(headline: str) -> str:
    prompt = f"Think step by step, then classify this headline.\nHeadline: {headline}"
```

...

Deliverables

- classifier.py with all three functions
- Printed accuracy table and per-approach token cost report